

ALIREZA MONTAJAB

B.Sc. Candidate in Electrical Engineering – Digital Electronics

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RESEARCH INTERESTS

Embedded and real-time systems • closed-loop control on resource-constrained hardware • sensor interfacing and vision-based feedback • digital system and datapath design

EDUCATION

B.Sc. in Electrical Engineering

Sept 2023 – Expected June 2027

Shahid Beheshti University, Tehran, Iran • ranked #741–750 worldwide and #214 in Asia (QS World University Rankings 2026)

- Specialisation: Digital Electronics (selected February 2026) • Cumulative GPA: 16.50/20 • Major GPA: 16.00/20
- **Senior Design Project** (in progress, Fall 2026 – Spring 2027): [topic], supervised by Dr. [name]
- **Relevant coursework:** Digital Systems I & II • Logic Circuits • Microprocessors • Electronics I–III • Signals & Systems • Linear Control Systems • Fundamentals of Mechatronics

ACADEMIC PROJECTS

Magnetic Levitation System with Visual Feedback

Spring 2026

Fundamentals of Mechatronics • Supervisor: Dr. Nouri-Manzar • Individual project

- Stabilised an equilibrium that is inherently unstable under Earnshaw's theorem, suspending a steel ball beneath an electromagnet in closed loop
- Replaced conventional IR and Hall-effect position sensing with an ESP32-CAM, extracting ball height from the centroid of dark pixels in 96×96 grayscale frames on the same microcontroller that ran the controller
- Tuned a PID loop (K_p 2.5, K_i 0.5, K_d 55) and constrained PWM output to a narrow 110–140 band, the large derivative term compensating for how quickly the ball falls
- Built the power stage around a logic-level IRLZ44N MOSFET, 1N4007 flyback diode and a hand-wound 220-turn coil; diagnosed gate-drive, common-ground and diode-polarity faults during bring-up

Water Level and Temperature Control System

Fall 2025

Digital Systems II final project • Supervisor: Dr. Asharioun • Team of 3; responsible for the complete firmware

- Regulated water level and temperature in a 50 cm tank from bare-metal STM32F103C8T6 firmware, written without third-party sensor libraries as the assignment required
- Measured level by timing the HC-SR04 echo pulse directly through TIM2 input capture and interrupt handling rather than a library driver
- Drove the inlet pump from a proportional controller over a 100 Hz PWM channel while a second loop read an LM35 through the ADC and held a heating element at 60 °C
- Built a menu-driven 16×4 LCD and 4×4 keypad interface with EXTI navigation, range-validated setpoint entry, RTC clock setting, UART confirmation and a low-temperature alarm

8-Bit Arithmetic Datapath in 74-Series Logic

Spring 2025

Digital Systems I final project • Supervisor: Dr. Farhad Pouladi • Individual project

- Designed and simulated an 8-bit arithmetic datapath in Proteus built entirely from discrete 74LS TTL devices, integrating arithmetic, data selection, decoding and storage into one clocked path
- Cascaded two 74LS283 carry-look-ahead adders into an 8-bit adder with 74LS86 XOR gates for bitwise comparison and parity generation
- Selected operand versus sum through 74LS157 multiplexers, routed output channels with a 74LS139 decoder, and stored results in 74LS194 bidirectional shift registers supporting parallel load, left/right shift and hold

ADDITIONAL COURSEWORK PROJECTS

- **Wi-Fi Controlled Smart Lighting Prototype** (Digital Systems II elective, Fall 2025, team of 3; led development) — ESP32 firmware in C++ hosting a standalone Wi-Fi access point and embedded HTTP server, with a responsive web interface and REST-style endpoints returning JSON device state
- **Discrete H-Bridge Motor Driver PCB** (Electronics II, Spring 2025) — laid out a four-transistor bidirectional DC motor driver in Altium Designer with flyback protection diodes, base current-limiting resistors, a DPDT direction switch and screw terminals
- **Transmission Line Parameter Calculator** (Power System Analysis I, Fall 2025) — built a MATLAB tool computing per-kilometre inductance and capacitance for transposed three-phase lines with arbitrary bundle geometry, using GMD and bundle-equivalent GMR with the $0.7788 r$ correction

- **Fourth-Order Op-Amp RLC Network Analysis** (*Electrical Circuits I, Spring 2024, team of 4*) — derived closed-form zero-input, step and impulse responses, natural frequencies, a state-space model and driving-point impedance, verifying every analytical result against LTspice transient and AC simulation

PROFESSIONAL EXPERIENCE

Tensor Intelligent Machines

Tehran, Iran

Robotics company at Sharif University Science & Technology Park, developing AI-driven autonomous parcel-sorting robots

Embedded Hardware Engineer Engineering Intern

June 2026 – Present
Sept 2025 – May 2026

Handheld Barcode Terminal — ESP32-S3, ESP-IDF, FreeRTOS

- Enumerated a commercial USB barcode scanner through the ESP-IDF USB Host HID stack as a boot-protocol keyboard, translating raw HID scancodes into ASCII and assembling complete barcode strings on terminator characters
- Partitioned the application across core-pinned FreeRTOS tasks communicating through queues, with a mutex guarding shared access to an SSD1306 OLED that reports battery level, channel selection, running scan count and the decoded barcode
- Added Li-ion monitoring from a 10-sample averaged ADC reading across a 2:1 divider mapped over the 3.0–4.2 V range, and designed the carrier PCB in Altium Designer

RFID Reader Board — Altium Designer

- Designed a surface-mount reader board around an AOD4185 P-channel load switch, BC817 driver transistors, an 18 V zener transient clamp, buzzer and four-button interface, with antenna and clock/data interfaces to the reader module
- Produced a fully sourced bill of materials with manufacturer part numbers and distributor pricing, and wrote serial capture firmware to decode raw module output during bring-up

Ultrasonic Ranging Module — STM32G030K6, Cortex-M0+

- Implemented HC-SR04 distance measurement on a 1 MHz TIM1 time base for microsecond echo timing, handling 16-bit counter overflow and adding timeout guards that fail safe when no echo returns

TEACHING EXPERIENCE

Instructor — Altium Designer PCB Design Workshop

Mechatronics Student Association, Shahid Beheshti University

- Designed and delivered a hands-on workshop for undergraduate students covering schematic capture, component library and footprint management, board layout and design-rule checking

Teaching Assistant — Electrical Circuits

Department of Electrical Engineering, Shahid Beheshti University

- Ran weekly problem-solving sessions, graded assignments and held office hours for the combined Circuits I & II course

TECHNICAL SKILLS

- **Embedded & Firmware:** C, C++, Embedded C, STM32 (HAL, CubeMX, CubeIDE), ESP32 / ESP32-S3 (ESP-IDF, Arduino), ARM Cortex-M0+/M3, FreeRTOS, bare-metal peripheral programming
- **Interfaces & Peripherals:** USB HID Host, UART, I²C, SPI, PWM, ADC, timers, input capture, external interrupts, RTC, DMA
- **Hardware & PCB:** Altium Designer, SMD schematic capture and layout, component libraries and footprints, BOM sourcing, MOSFET power stages, motor drivers, sensor interfacing (HC-SR04, LM35, RFID, camera modules)
- **Digital Design:** 74-series TTL logic, combinational and sequential design, adders, multiplexers, decoders, shift registers, finite state machines, datapath design
- **Control & Algorithms:** PID design and tuning, proportional control, closed-loop feedback systems, real-time image processing
- **Simulation & Tools:** Proteus (mixed-signal), LTspice, MATLAB / Simulink, oscilloscope, logic analyser, hardware debugging, Git, Linux, LaTeX

HONORS & AWARDS

- **Full Tuition Scholarship**, 2023–2027 — admitted to a fully state-funded place at Shahid Beheshti University on the basis of national entrance examination rank; covers the entire B.Sc.
- **National University Entrance Examination (Konkur)**, 2023 — ranked in the top 1% nationally
- **Languages:** English — IELTS Academic, Overall 7.0 • Persian — Native